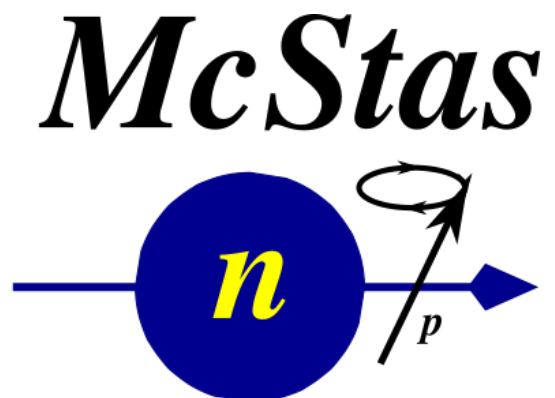




Simulating Polarized Neutron Scattering Experiments
and Equipment with McStas

Peter Willendrup, DTU Physics & ESS DMSC

Presentation content

- Polarisation in McStas - how is it done
 - Polarised monitors
 - Precession algorithm
- Polarising by spin up / down reflectivities
- An overview of polarisation comps
- An overview of polarisation instrument examples
- A quick look at 3 of these
- Words of warning

McStas “particle” model

Neutron ray/package:

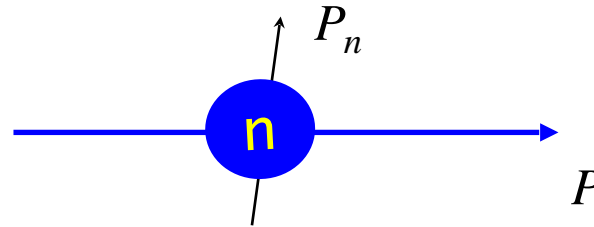
Weight: (p) # neutrons left in the package

Position: (x, y, z)

Velocity: (v_x , v_y , v_z)

Polarization: (s_x , s_y , s_z)

Time: (t)



“Within one ray”

$$P_n = \frac{1}{p_n} \sum_i^p P_{i,n}; \quad n = \text{raynumber}$$

“Over the beam”

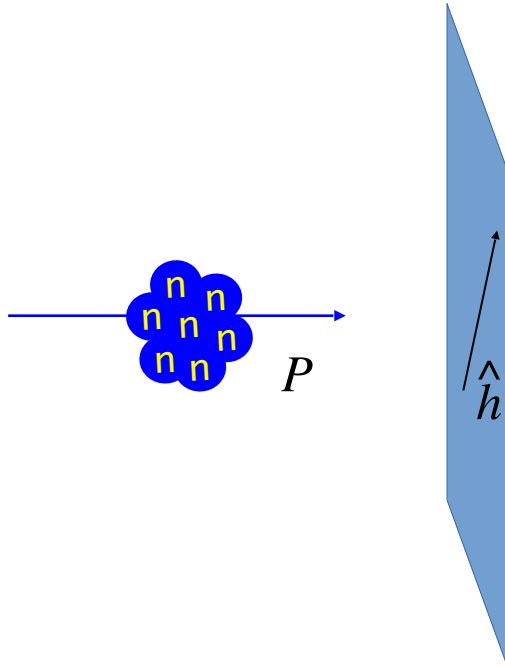
$$P = \frac{1}{N} \sum_{n=0}^N P_n$$

$$P_{i,n} = 2 \left(\langle \hat{s}_{x,i} \rangle \hat{i}_{x,i} + \langle \hat{s}_{y,i} \rangle \hat{i}_{y,i} + \langle \hat{s}_{z,i} \rangle \hat{i}_{z,i} \right)$$

From G. Williams: “Polarized neutrons”, Oxford Science Publ., 1988

McStas detectors/monitors

Monitoring: How and What do we monitor?

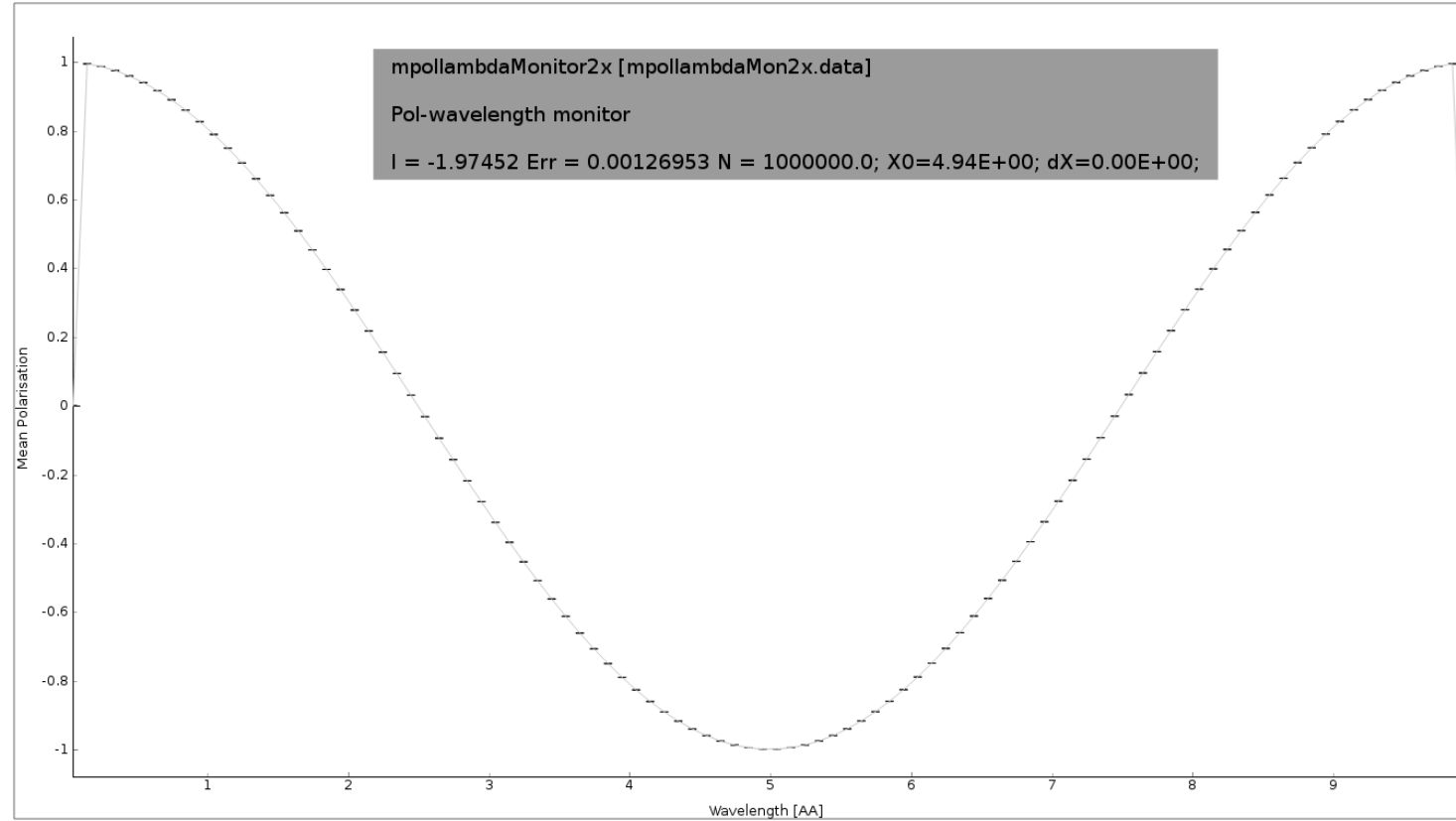
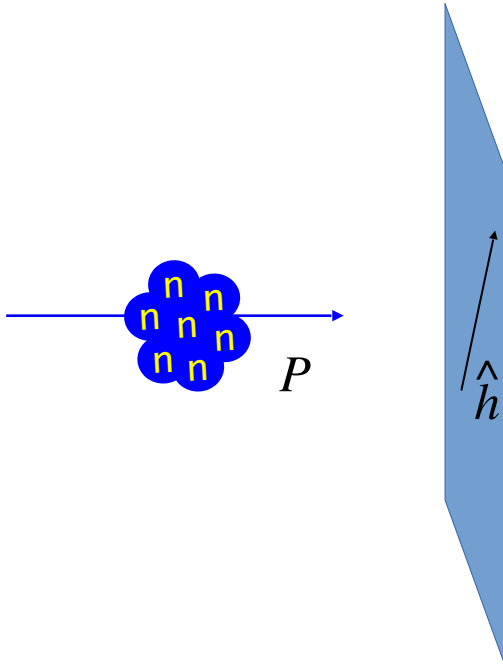


$$P_{\hat{h}} = \frac{\sum_{n=0}^N p_n P_n \cdot \hat{h}}{\sum_n p_n}$$

(Statistic-) weighted polarisation along chosen monitoring direction

McStas detectors/monitors

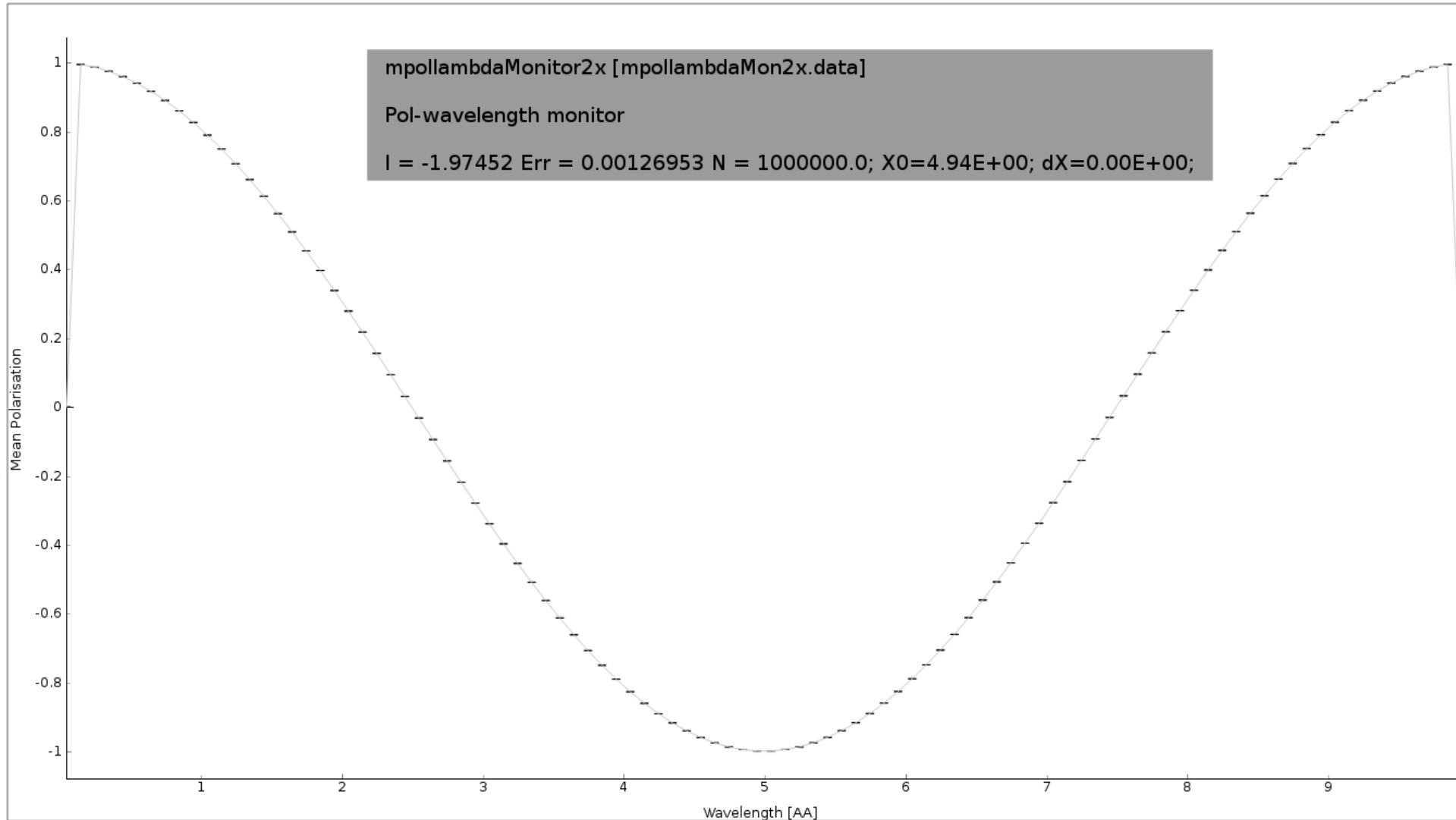
Monitoring: How and What do we monitor?



E.g. polarisation along $\hat{h}$ as fct. of wavelength

McStas detectors/monitors

Monitoring: How and What do we monitor?



Nota bene:
weighted
polarisation,
not intensity

Polarisation along $\hat{h}$ as fct. of wavelength

Polarization monitors

- Available monitors:
 - `Pol_monitor.comp`: 0D
 - `PolLambda_monitor.comp`: 2D
 - `PolTOF_monitor.comp`: 2D
 - `MeanPolLambda_monitor.comp`: 1D

McStas precession algorithm

- Magnetic fields in McStas
- The challenge:
 - Fast beam/ray transport: # $rays > 10^6$
 - Unknown magnetic field and field strength
 - >1 Magnet $\rightarrow$ nested fields.

McStas precession algorithm

```

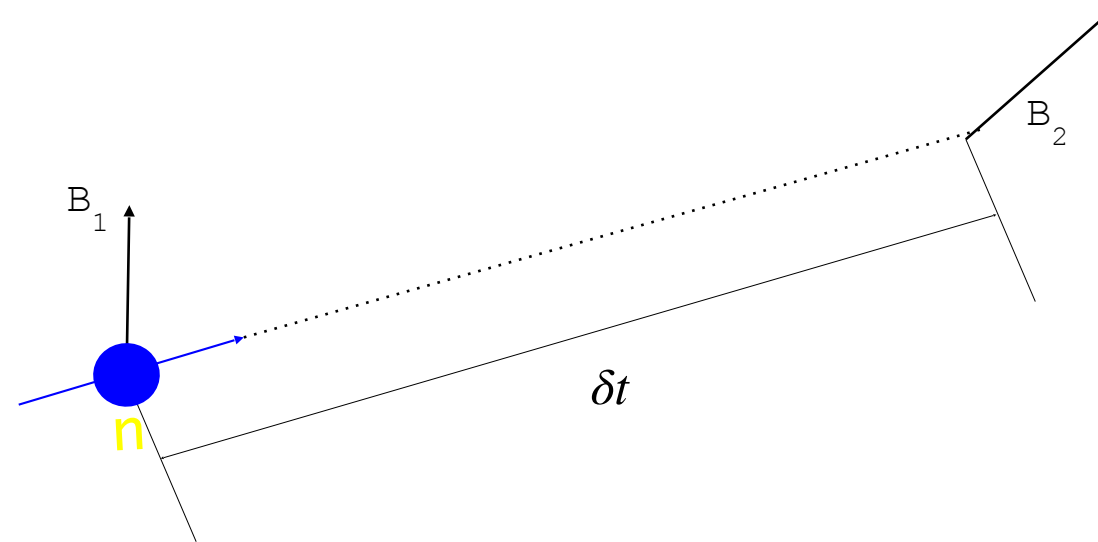
while  $n_t < t_{\text{target}}$  do
  store neutron;
  sample magnetic field:  $\mathbf{B}_1 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
  propagate neutron:  $\delta t (< \Delta t)$ ;
  sample magnetic field:  $\mathbf{B}_2 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
  while  $|\mathbf{B}_1 - \mathbf{B}_2| > \delta B_{\text{threshold}}$  do
    restore neutron;
     $\delta t := \delta t / 2$ ;
    propagate neutron:  $\delta t (< \Delta t)$ ;
    sample magnetic field:  $\mathbf{B}_1 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
  precess polarization:  $\mathbf{P}_n$  by  $\omega$  around  $\frac{\mathbf{B}_1 + \mathbf{B}_2}{2}$ ;

```

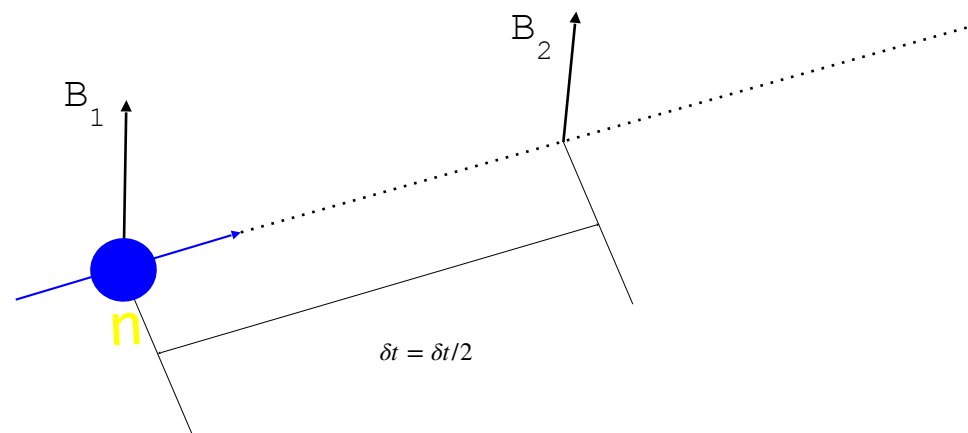
Algorithm 1: SimpleNumMagnetPrecession: Simplistic algorithm for tracking polarization of a Monte-Carlo neutron in a magnetic field. The neutron's state is stored as a position (n_x, n_y, n_z) , a velocity $\mathbf{v}$, time n_t , and polarization vector $\mathbf{P}_n$.

From: Knudsen et.al., J. Neutron Research, 2014

McStas precession algorithm



McStas precession algorithm



McStas precession algorithm

```

while  $n_t < t_{\text{target}}$  do
  store neutron;
  sample magnetic field:  $\mathbf{B}_1 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
  propagate neutron:  $\delta t (< \Delta t)$ ;
  sample magnetic field:  $\mathbf{B}_2 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
  while  $|\mathbf{B}_1 - \mathbf{B}_2| > \delta B_{\text{threshold}}$  do
    restore neutron;
     $\delta t := \delta t / 2$ ;
    propagate neutron:  $\delta t (< \Delta t)$ ;
    sample magnetic field:  $\mathbf{B}_1 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
  precess polarization:  $\mathbf{P}_n$  by  $\omega$  around  $\frac{\mathbf{B}_1 + \mathbf{B}_2}{2}$ ;

```

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From: Knudsen et.al., J. Neutron Research, 2014

McStas precession algorithm

```

while  $n_t < t_{\text{target}}$  do
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  sample magnetic field:  $\mathbf{B}_2 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
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    sample magnetic field:  $\mathbf{B}_1 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
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```

Algorithm 1: SimpleNumMagnetPrecession: Simplistic algorithm for tracking polarization of a Monte-Carlo neutron in a magnetic field. The neutron's state is stored as a position (n_x, n_y, n_z) , a velocity $\mathbf{v}$, time n_t , and polarization vector $\mathbf{P}_n$.

From: Knudsen et.al., J. Neutron Research, 2014

McStas precession algorithm

```

while  $n_t < t_{\text{target}}$  do
  store neutron;
  sample magnetic field:  $\mathbf{B}_1 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
  propagate neutron:  $\delta t (< \Delta t)$ ;
  sample magnetic field:  $\mathbf{B}_2 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
  while  $|\mathbf{B}_1 - \mathbf{B}_2| > \delta B_{\text{threshold}}$  do
    restore neutron;
     $\delta t := \delta t / 2$ ;
    propagate neutron:  $\delta t (< \Delta t)$ ;
    sample magnetic field:  $\mathbf{B}_1 = \mathbf{B}(n_x, n_y, n_z, n_t)$ ;
  precess polarization:  $\mathbf{P}_n$  by  $\omega$  and  $\mathbf{B}_1$ 

```

Algorithm 1: SimpleNumMagnetPrecession. Precessing polarization of a Monte-Carlo neutron state is stored as a position (n_x, n_y, n_z) and a vector $\mathbf{P}_n$.

From: Knudsen et.al., J. Neutron Research

```

/*Threshold below which two magnetic fields are considered to be
 * in the same direction.*/
#ifndef mc_pol_angular_accuracy
#define mc_pol_angular_accuracy (1.0*DEG2RAD)
#endif

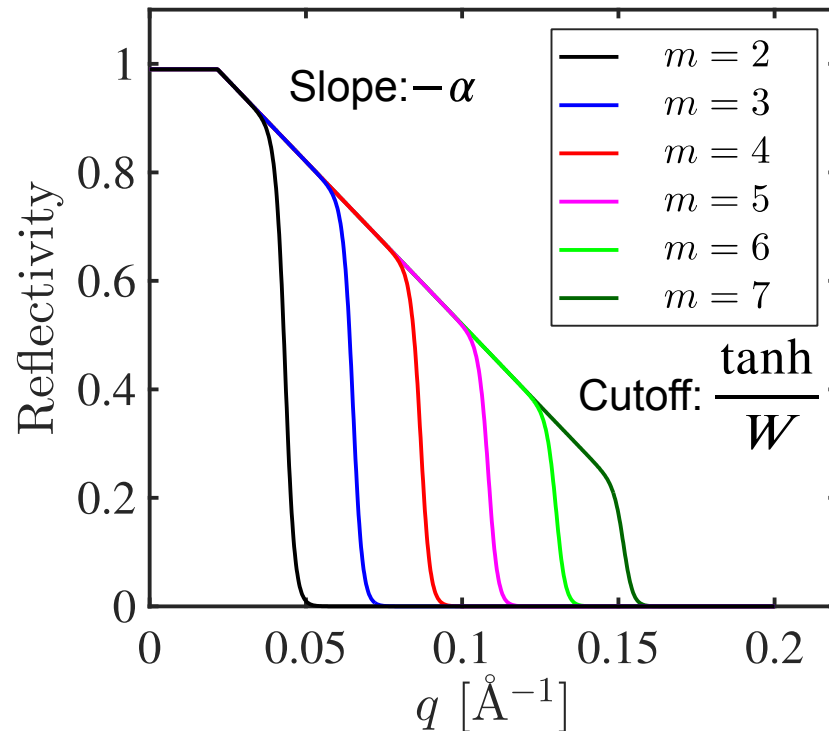
/*The maximal timestep taken by neutrons in a const field*/
#ifndef mc_pol_initial_timestep
#define mc_pol_initial_timestep 1e-5;
#endif

```

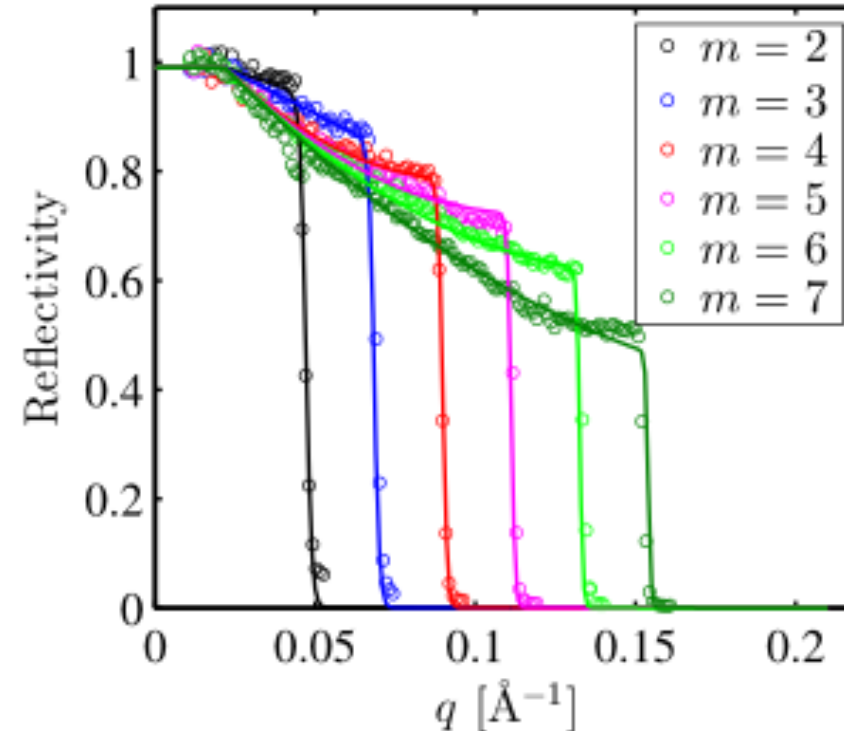
Reminder: Reflectivity curves in McStas

$$R(q) = \begin{cases} R_0 & \text{if } q < q_c \\ R_0(1 - \tanh((q - mq_c)/W))(1 - \alpha(q - q_c))/2 & \text{otherwise} \end{cases}$$

McStas standard model



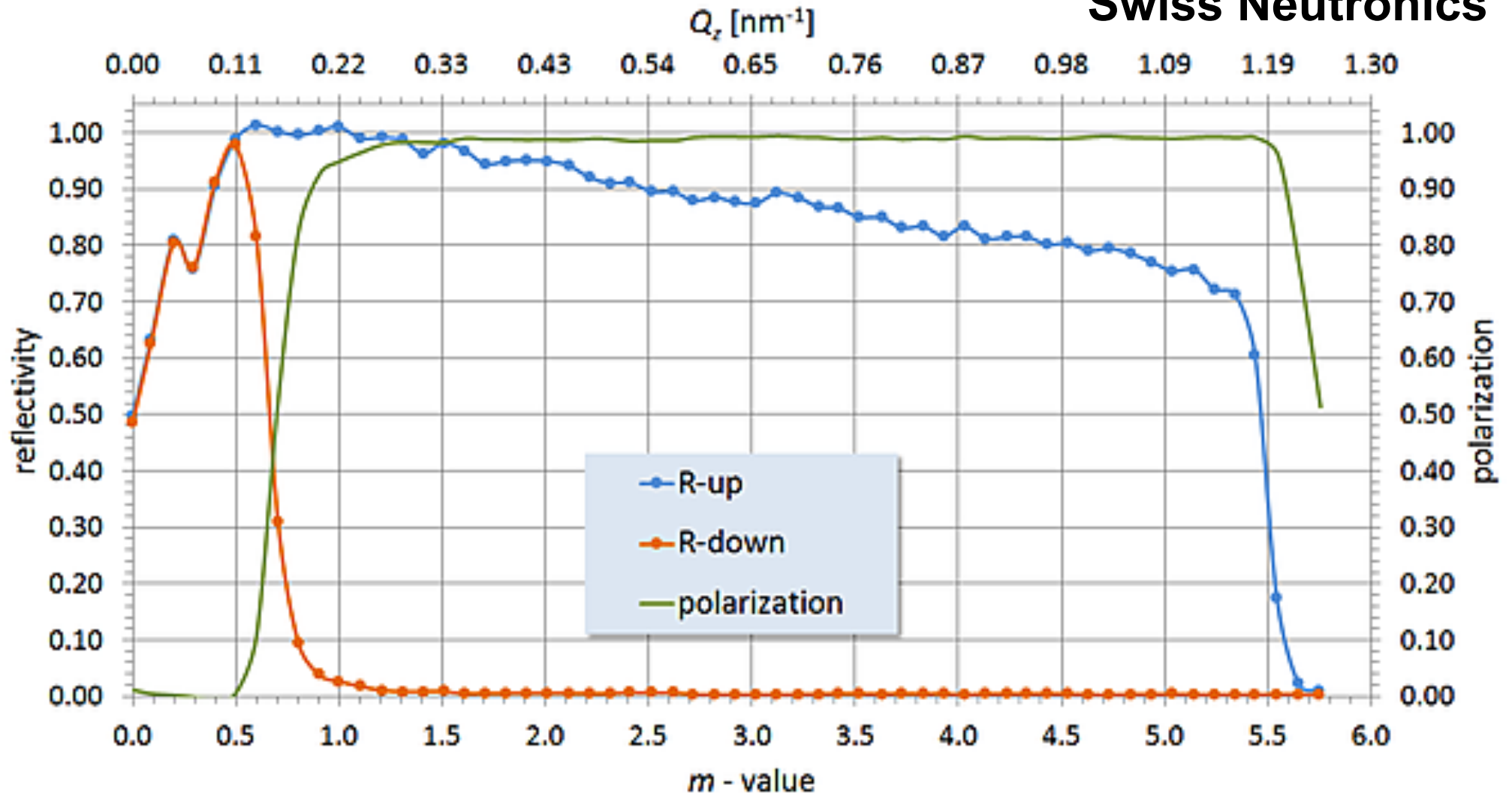
McStas fitted model



$\alpha = 0$
 $W = 0$
 Only m matters
 Better mirrors
 available today

State-of-the-art Polarising Supermirrors (2019)

Swiss Neutronics



Depending on the component, several underlying “reflectivity” functions may be used:

That is: // The McStas standard analytic parametrization of the reflectivity

In McStas void StdReflecFunc(double q, double *par, double *r);

COMPO // Reflectivity from a table using the routines in read_table-lib.

rDown void TableReflecFunc(double q, t_Table *par, double *r);

rUpPar

zwidth // The McStas standard analytic parametrization of the reflectivity for

AT (0, 0, // double-side coated supermirror.

void StdDoubleReflecFunc(double q, double *par, double *r);

// “Extended” reflectivity fct. with 2nd order ‘beta’ parameter in mc_pol_par

void ExtendedReflecFunc(double mc_pol_q, double *mc_pol_par, double *mc_pol_r);

McStas polarization components

Magnetic fields:

- `Pol_FieldBox.comp`
- `Pol_Bfield.comp`
- `Pol_Bfield_stop.comp`
- `Pol_triafield.comp`
- `(Pol_tabled_field 3.x)`

Monitors:

- `Pol_monitor.comp`
- `MeanPolLambda_monitor.comp`
- `PolLambda_monitor.comp`
- `PolTOF_monitor.comp`

Contrib:

- `Foil_flipper_magnet.comp`

Optics:

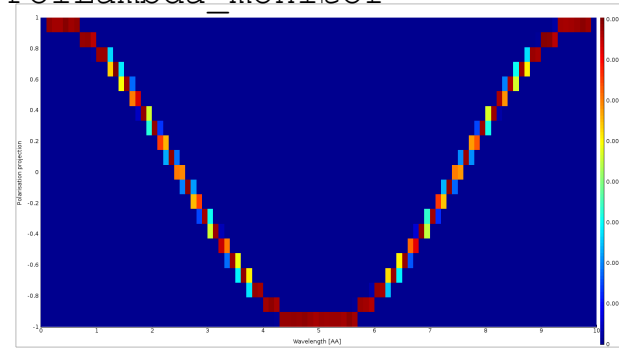
- `Monochromator_pol.comp`
- `Pol_bender.comp`
- `Pol_guide_mirror.comp`
- `Pol_guide_vmirror.comp`
- `Pol_mirror.comp`
- `Transmission_polarisatorABSnT.comp`
- `Pol_bender_tapering.comp`

Idealized components:

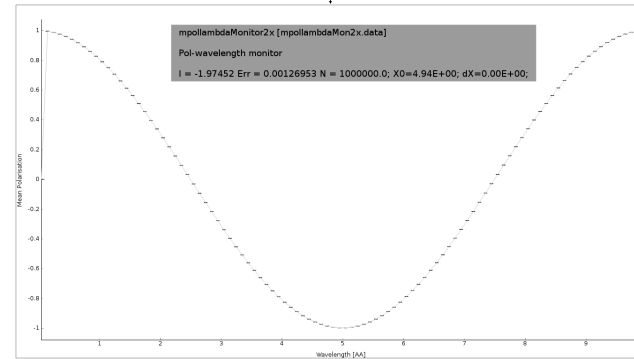
- `PolAnalyser_ideal.comp`
- `Pol_SF_ideal.comp`
- `Pol_pi_2_rotator.comp`
- `Set_pol.comp`

McStas polarization monitors

PolLambda_monitor



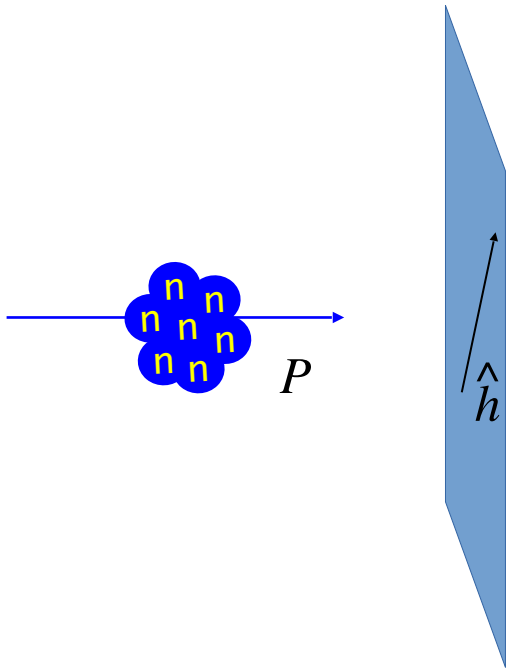
MeanPolLambda_monitor



Monitors

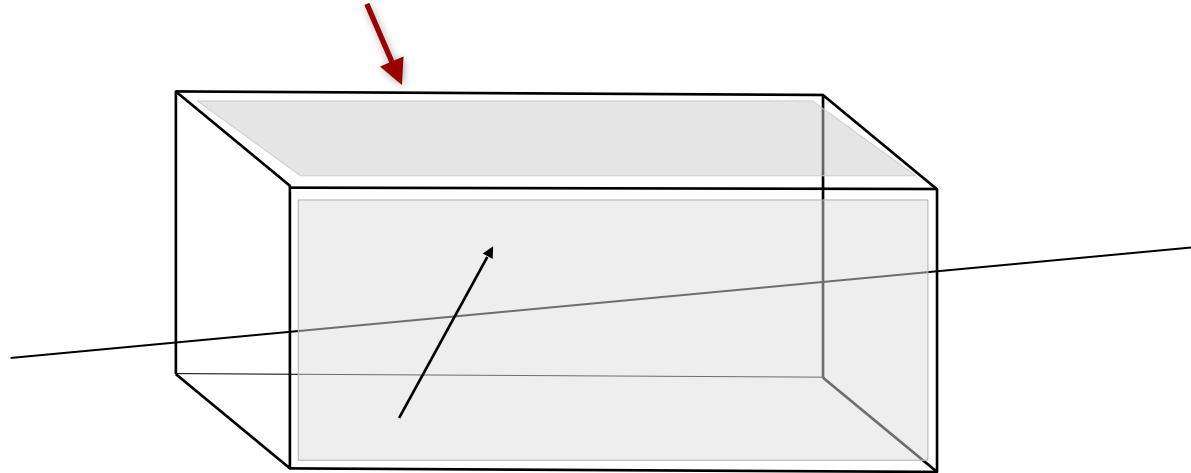
Pol_monitor

$$P \parallel (m_x, m_y, m_z) = 0.87$$



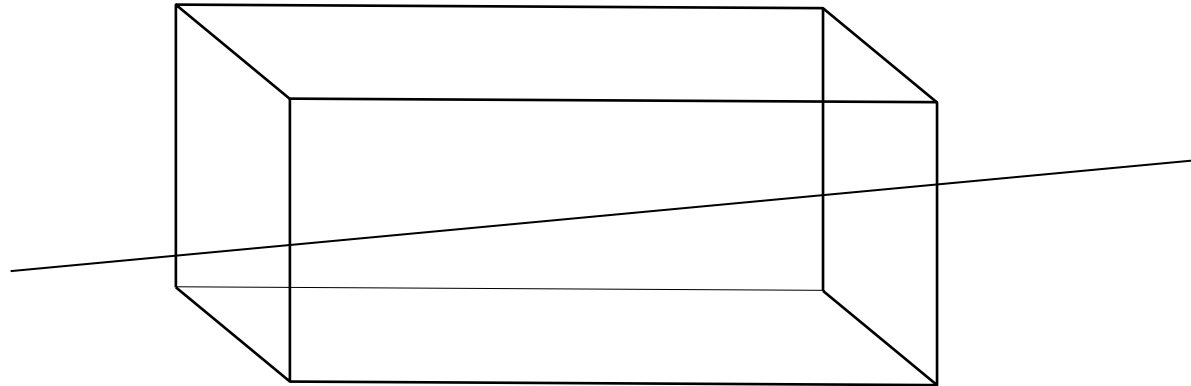
McStas magnetic fields

- `Pol_constBfield.comp`
- Single constant Magnetic field in a “box”.
- - user may specify a wavelength to flip.
- “blocking walls”



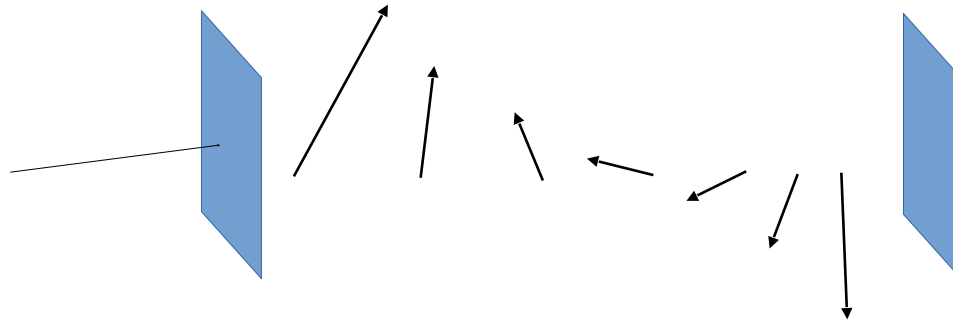
McStas magnetic fields

- `Pol_FieldBox.comp`
- Single Magnetic field in a “box”
- Constant or tabled magnetic fields



McStas magnetic fields

- `Pol_Bfield.comp`
- `Pol_Bfield_stop.comp`
 - Entry/Exit contruction allows for nested magnetic field descriptions.
 - Any magnetic field through user supplied c-function
 - Tabled magnetic fields



Standard field types:

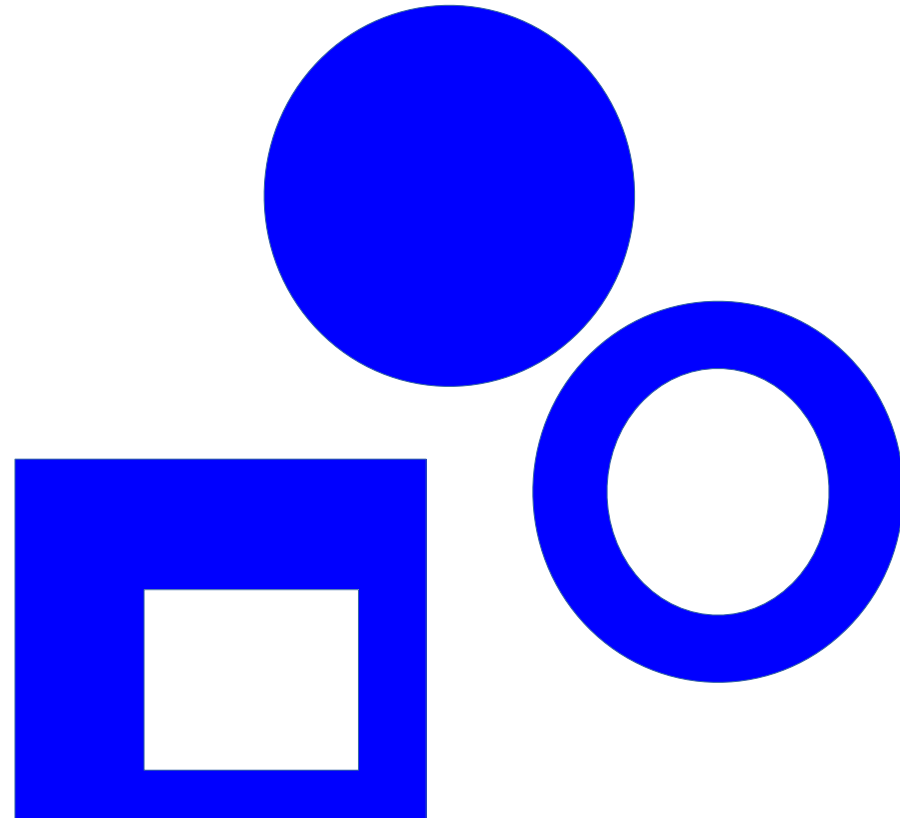
- Constant field
- Rotating field
- Gradient field
- “Majorana” type field

Plus user-defined fields (McStas 2.x only)

See `pol-lib.c` in `share/` and the examples

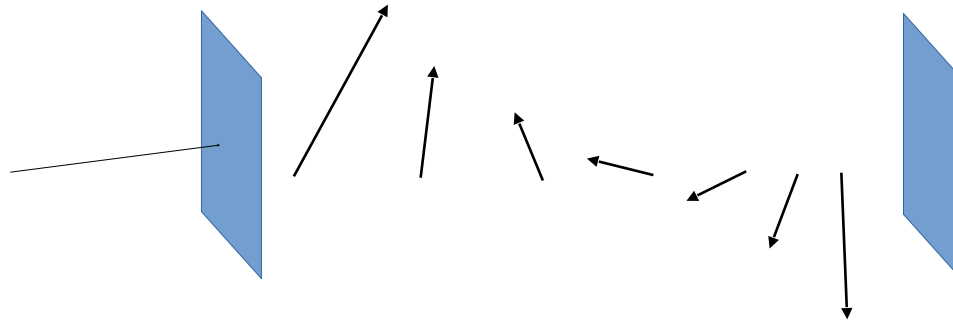
Windows can be many shapes

B-Fields: constant, functional, tabled, ... in more general shapes



McStas magnetic fields

- `Pol_Bfield.comp`
- `Pol_Bfield_stop.comp`
 - Entry/Exit contruction allows for nested magnetic field descriptions.
 - Any magnetic field through user supplied c-function
 - Tabled magnetic fields



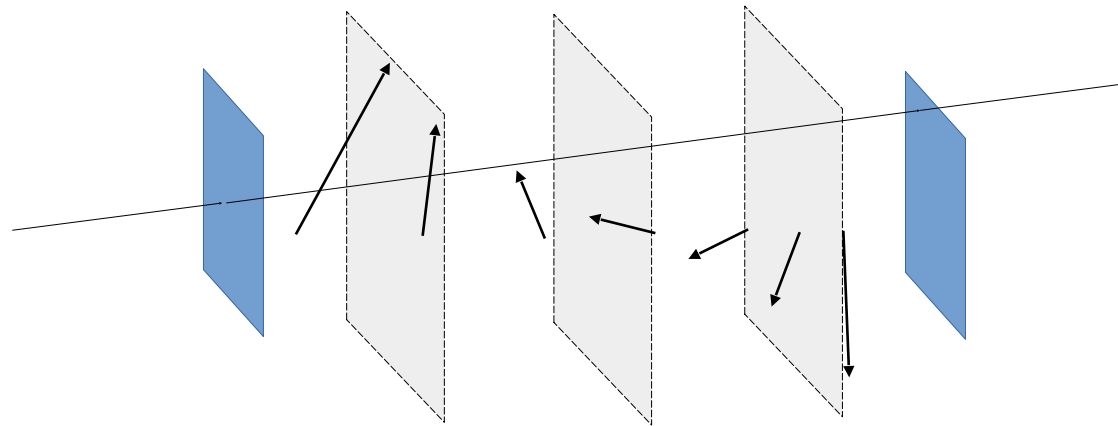
Standard field types:

- Constant field
- Rotating field
- Gradient field
- “Majorana” type field

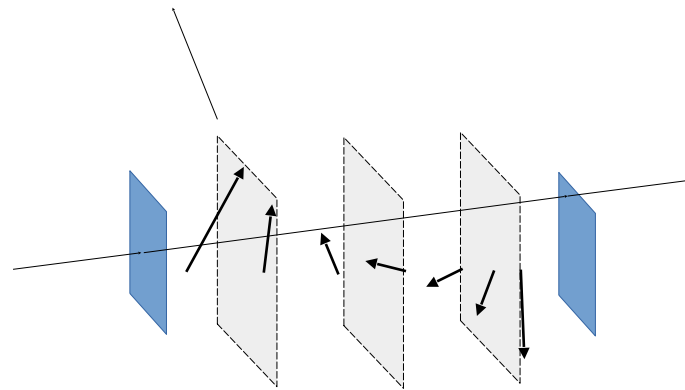
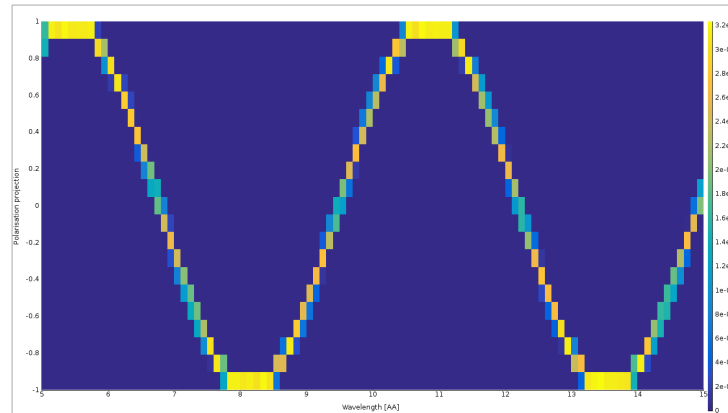
Plus user-defined fields (McStas 2.x only)

See `pol-lib.c` in `share/` and the examples

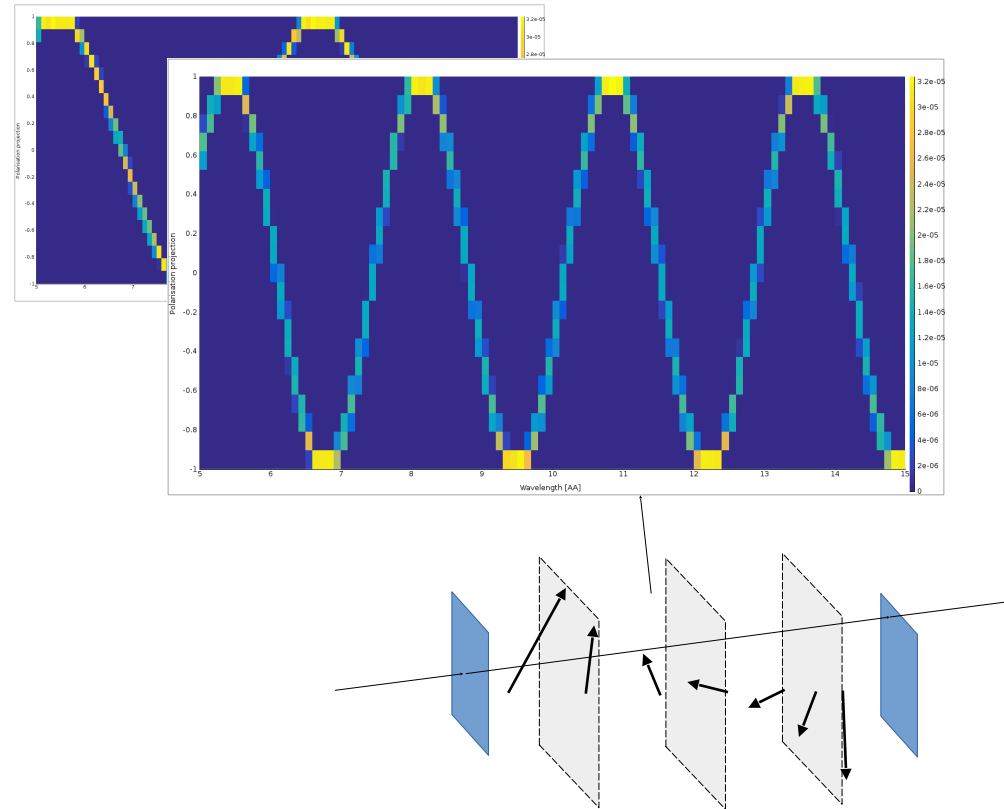
Pol_monitors along the way...



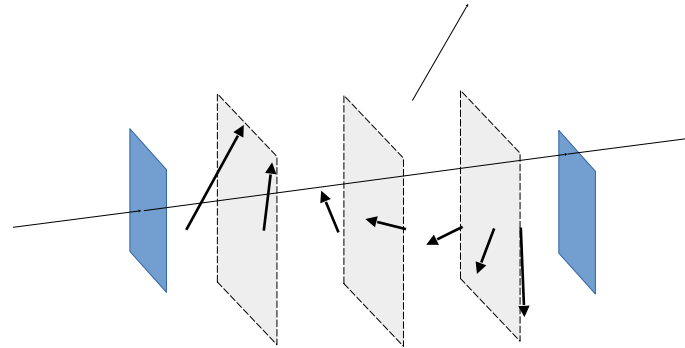
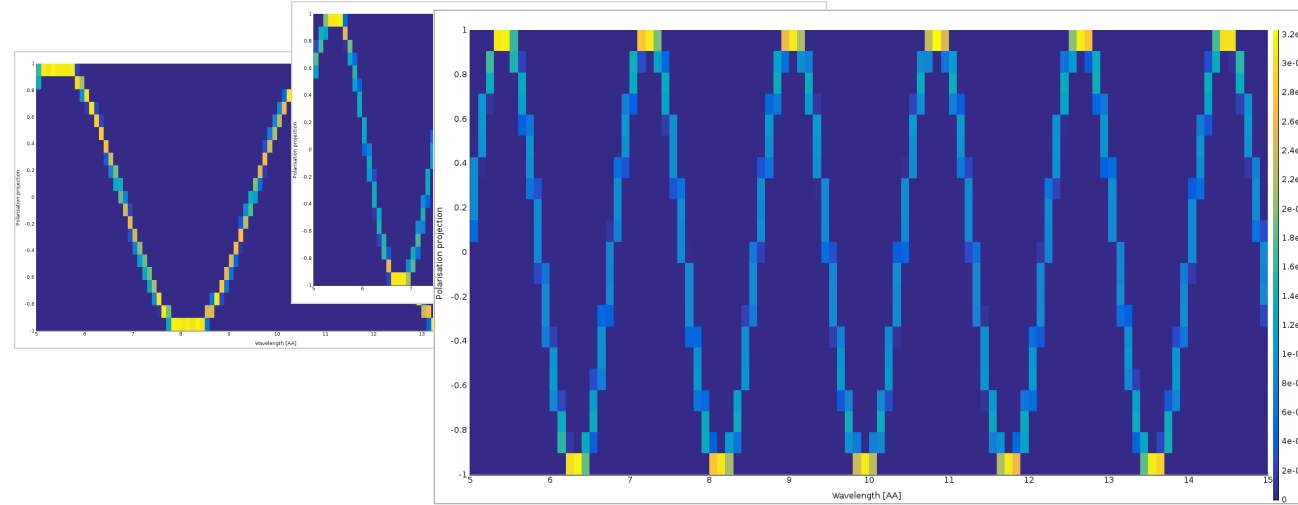
Pol_monitors along the way...



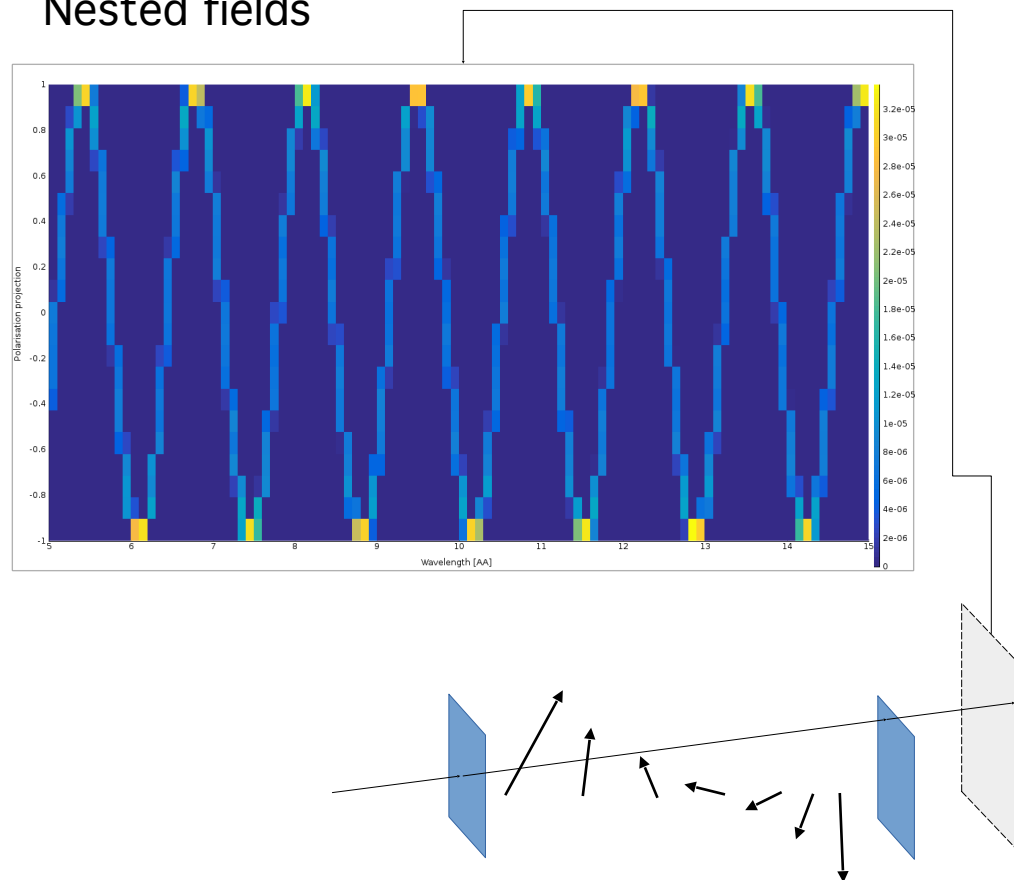
Pol_monitors along the way...



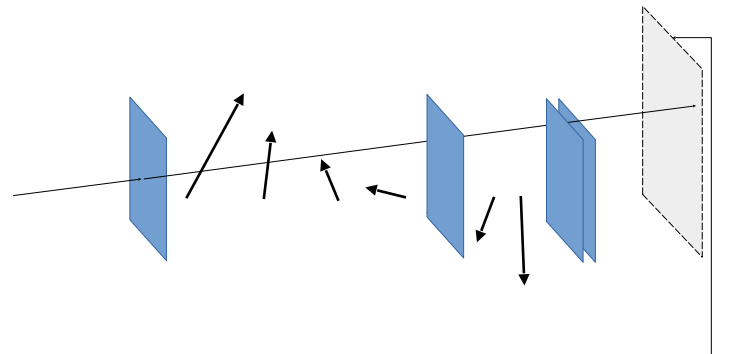
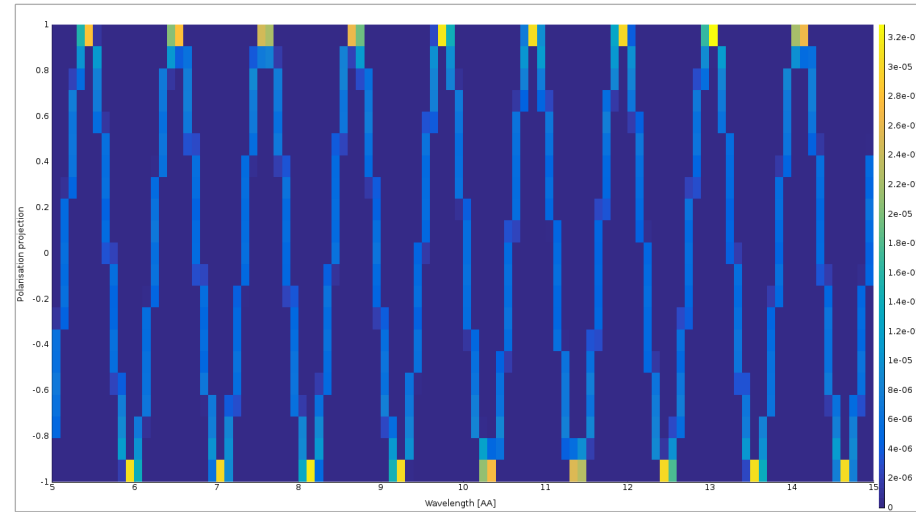
Pol_monitors along the way...



Nested fields



Nested fields



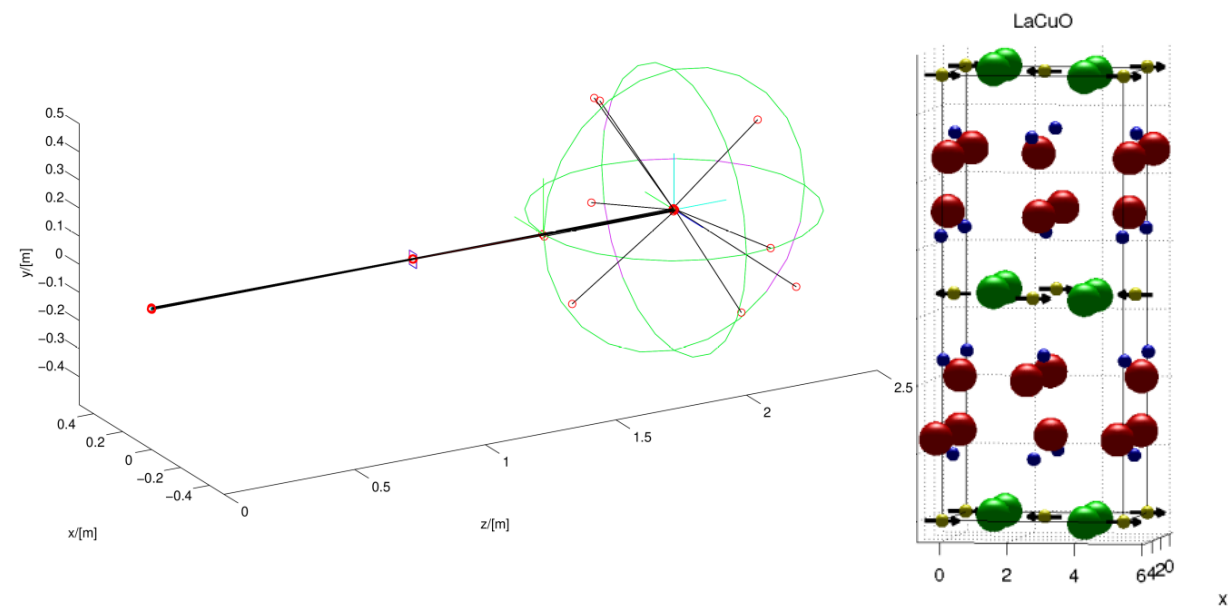
Simple McStas sample component

Incoherent.comp has SF / NSF solution

```
/* Polarisation part (1/3 NSF, 2/3 SF) */  
sx *= -1.0/3.0;  
sy *= -1.0/3.0;  
sz *= -1.0/3.0;  
  
SCATTER;
```

McStas sample component

Magnetic single crystal

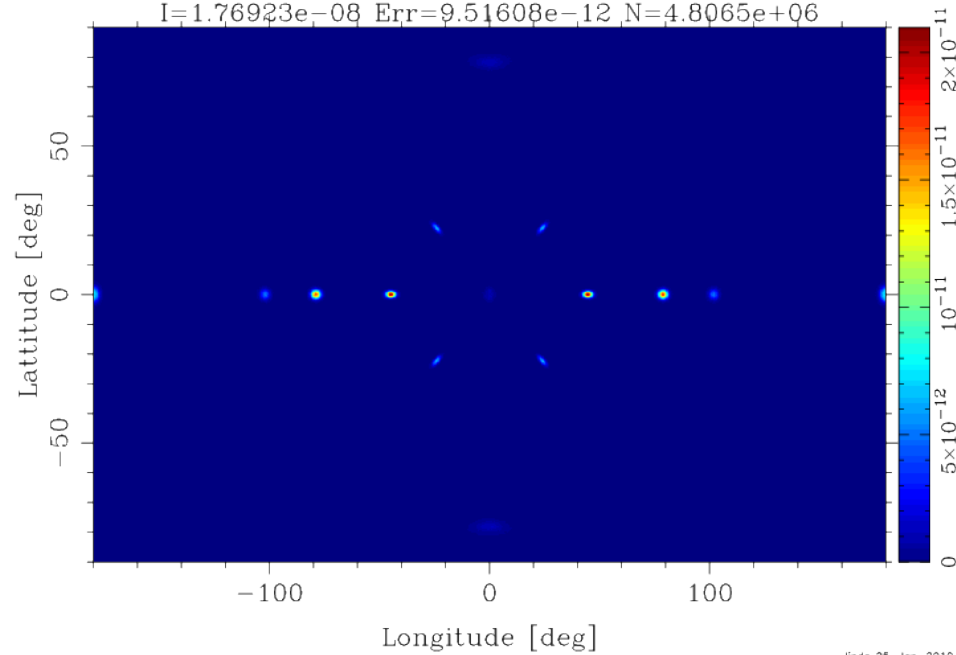


index	iontype	x	y	z	$b_{coh}[\text{fm}]$	g_S	S_x	S_y	S_z	g_L	L_x	L_y	L_z
1	Cu2+	0.5	0.5	0	7.718	2	0	-0.5	0	0	0	0	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

McStas sample component

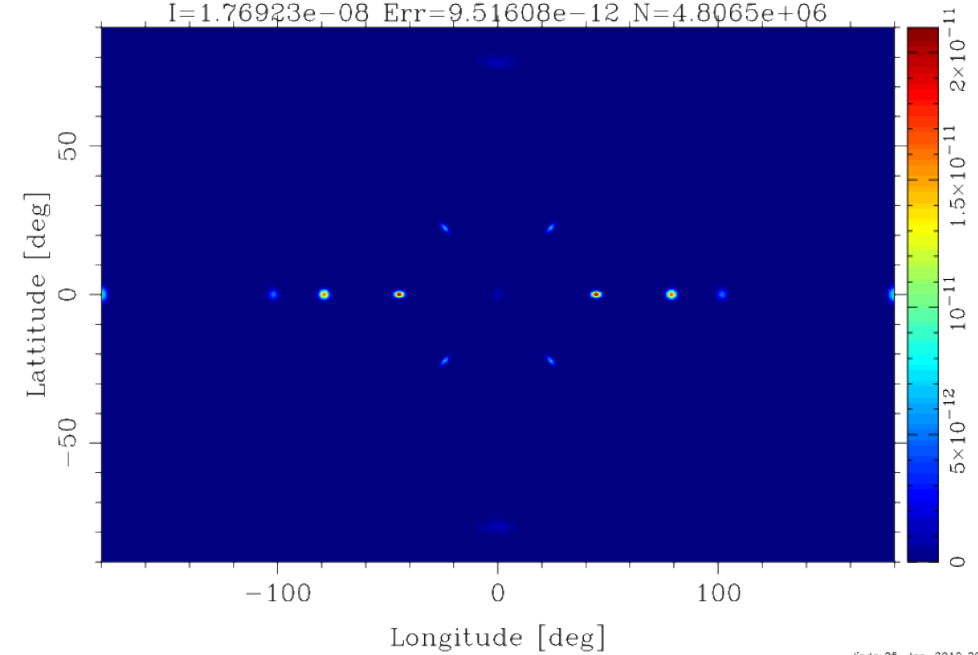
Magnetic single crystal – **Unpolarized** beam

4PImon_spinup [250110_SF_NSF_PX0_PY0_PZ0_1e10/PSD4PImon_spinup.
X0=-0.104202; dX=88.4169; Y0=0.105552; dY=25.2284;
I=1.76923e-08 Err=9.51608e-12 N=4.8065e+06



linda 25-Jan-2010 20:45

lmon_spindown [250110_SF_NSF_PX0_PY0_PZ0_1e10/PSD4PImon_spindown
X0=-0.104202; dX=88.4169; Y0=0.105552; dY=25.2284;
I=1.76923e-08 Err=9.51608e-12 N=4.8065e+06

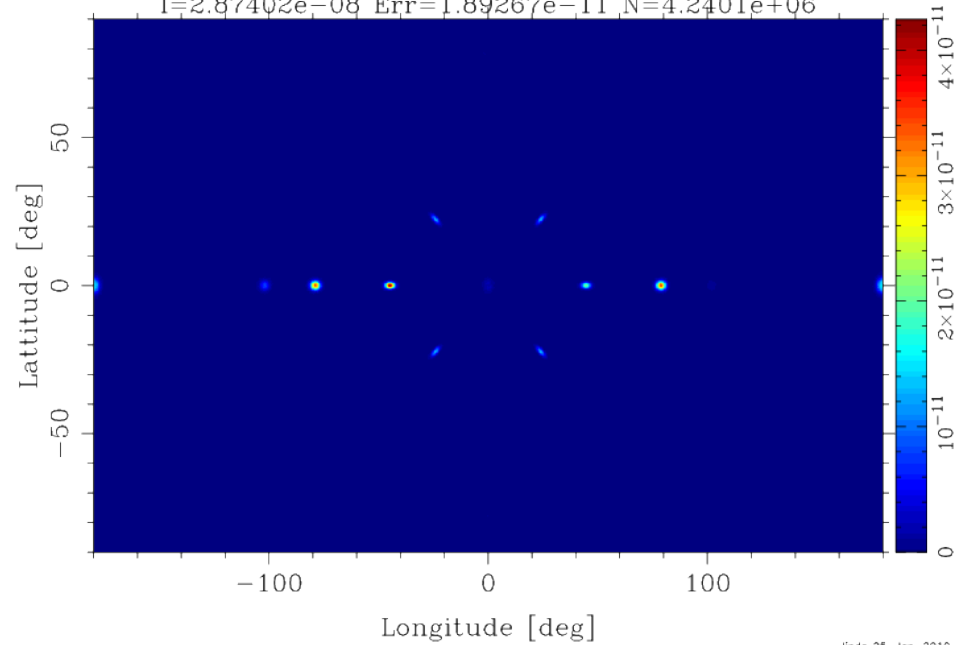


linda 25-Jan-2010 20:45

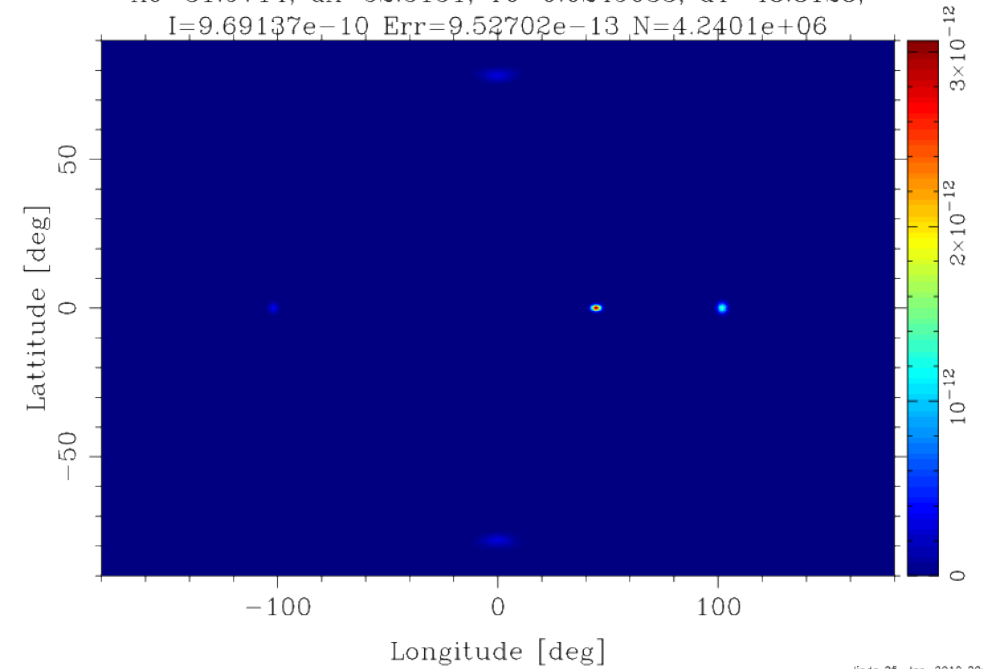
McStas sample component

Magnetic single crystal – **Polarized** beam

n_spinup [250110_SF_NSF_PX-0.3800_PYO_PZ0.9249_1e10/PSD4Plmon_s]
X0=-7.4526; dX=93.2595; Y0=0.0952368; dY=15.1242;
I=2.87402e-08 Err=1.89267e-11 N=4.2401e+06



spindown [250110_SF_NSF_PX-0.3800_PYO_PZ0.9249_1e10/PSD4Plmon_s]
X0=31.9714; dX=52.5151; Y0=0.0249033; dY=48.8128;
I=9.69137e-10 Err=9.52702e-13 N=4.2401e+06



McStas sample component

Magnetic single crystal

The magnetic scattering cross-section for a sample with localised spin+orbital angular momentum $g\mathbf{J} = (g_S + g_L)\mathbf{J} = 2\mathbf{S} + \mathbf{L}$ is:

$$\frac{d^2\sigma}{d\Omega_f dE_f} = \frac{k_f}{k_i} \sum_{i,f} P(\lambda_i) \left| \langle \lambda_f | \sum_j e^{i\mathbf{Q} \cdot \mathbf{d}_j} U_j^{\sigma_i \sigma_f} | \lambda_i \rangle \right|^2 \delta(\hbar\omega + E_i - E_f)$$

where $|\lambda_i\rangle$ and $\langle\lambda_f|$ are the initial and final states of the sample with energies E_i and E_f respectively, $P(\lambda_i)$ is the distribution of initial states and

$$U_j^{\sigma_i \sigma_f} = \langle \sigma_f | b_j - m_j \mathbf{J}_{\perp j} \cdot \boldsymbol{\sigma} | \sigma_i \rangle$$

where $|\sigma_i\rangle$ and $\langle\sigma_f|$ are the initial and final spin states of the neutron, and $\boldsymbol{\sigma}$ are the Pauli spin matrices working on the neutron state.

From: G. Shirane et.al. , "Neutron Scattering with Triple-Axis Spectrometer",
Cambridge Univ. Press, 2002

McStas sample component

Magnetic single crystal

If $\mathbf{P} = P(\xi, \eta, \zeta) = P\hat{\zeta}$. Thus, the matrix elements of $U^{\sigma_i \sigma_f}$ can now be written

$$\begin{aligned}U^{++} &= b - mJ_{\perp\zeta} \\U^{--} &= b + mJ_{\perp\zeta} \\U^{+-} &= -m(J_{\perp\xi} + iJ_{\perp\eta}) \\U^{-+} &= -m(J_{\perp\xi} - iJ_{\perp\eta})\end{aligned}$$

where $m = \frac{r_0\gamma}{2}gf(\mathbf{Q})$ with r_0 the classical electron radius, $\gamma = 1.913$, g the Landé splitting factor and $f(\mathbf{Q})$ the magnetic form factor of a particular ion in the sample.

Example instruments:

`*Magnetic*.instr:`

`Test_Magnetic_Constant.instr`

`Test_Magnetic_Majorana.instr`

`Test_Magnetic_Rotation.instr`

`Test_Magnetic_Userdefined.instr (2.x only)`

`Test_single_magnetic_crystal.instr (2.x only)`

`SE*.instr:`

`SEMSANS_Delft.instr`

`SEMSANS_instrument.instr`

`SESANS_Delft.instr`

`SE_example.instr`

`SE_example2.instr`

`*Pol*.instr:`

`Test_Pol_Bender.instr`

`Test_Pol_Bender_Vs_Guide_Curved.instr`

`Test_Pol_FieldBox.instr`

`Test_Pol_Guide_Vmirror.instr`

`Test_Pol_Guide_mirror.instr`

`Test_Pol_MSf.instr`

`Test_Pol_Mirror.instr`

`Test_Pol_SF_ideal.instr`

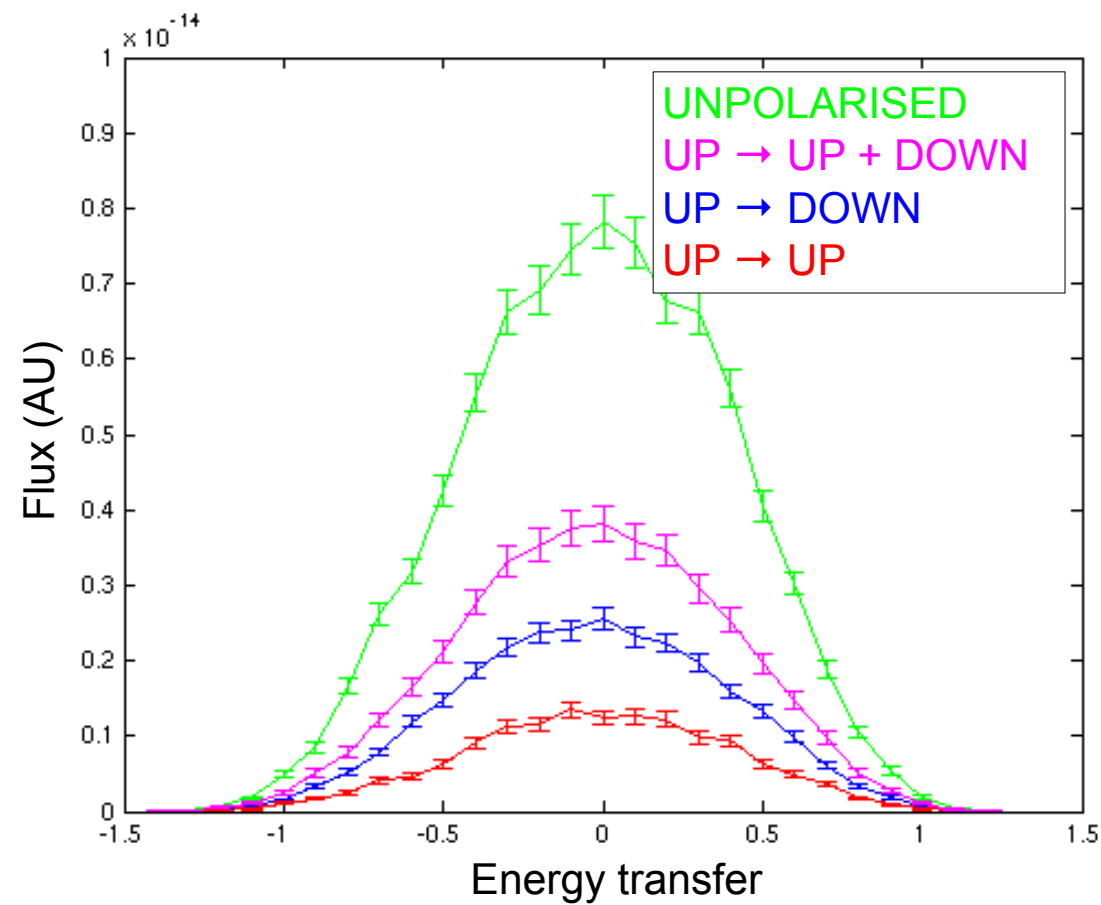
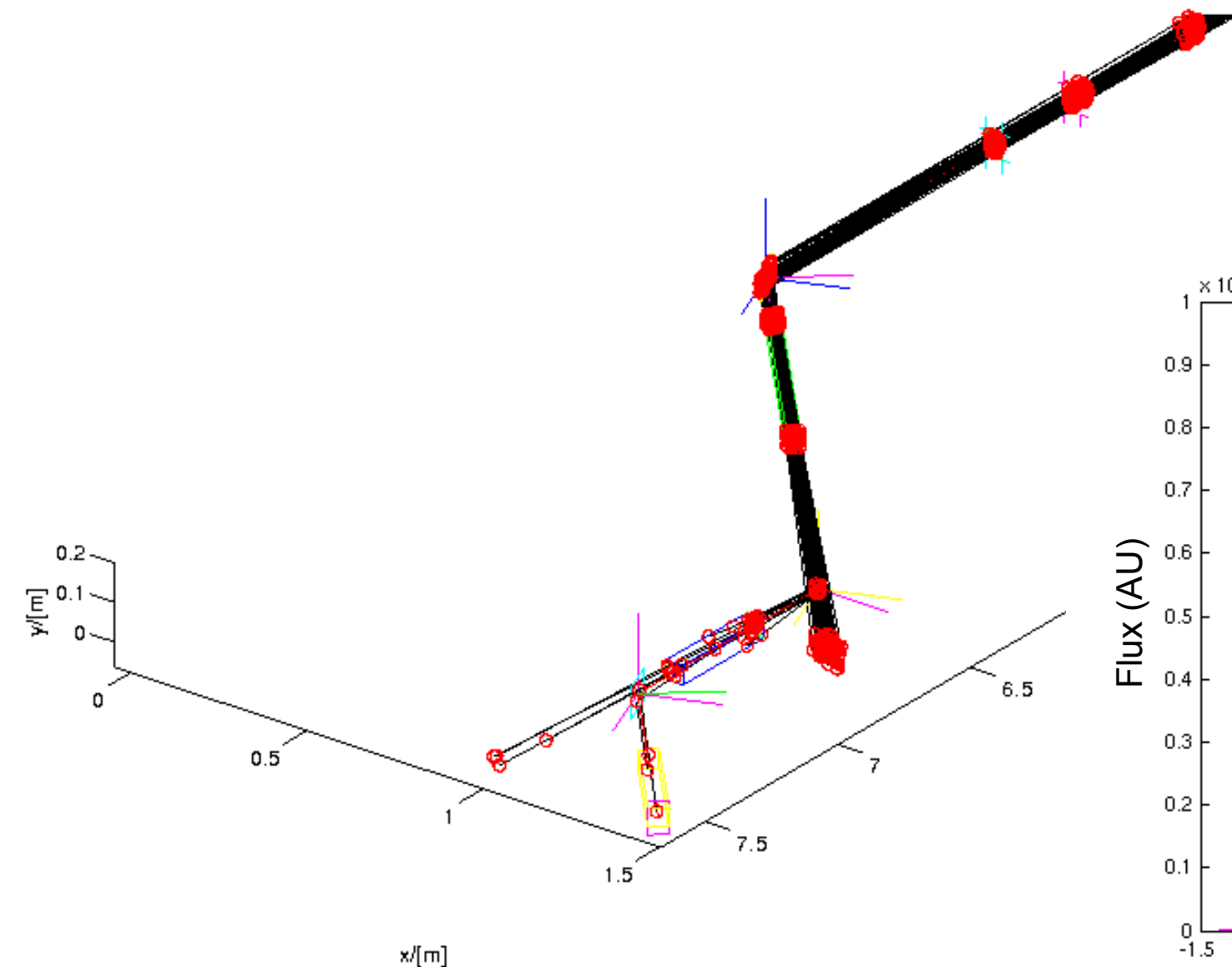
`Test_Pol_Set.instr`

`Test_Pol_Tabled.instr`

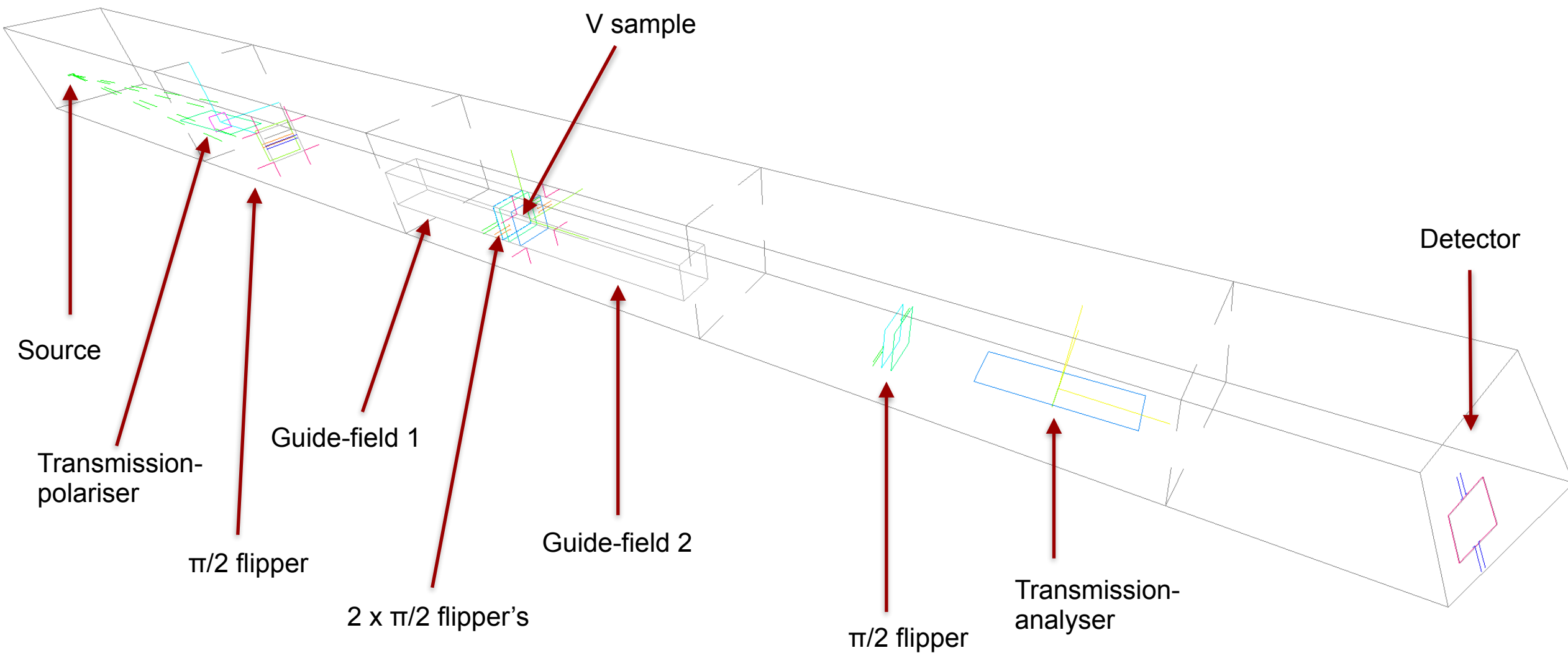
`Test_Pol_TripleAxis.instr`

A copy of the files from McStas 3.5.x is placed in the 'instrs' folder

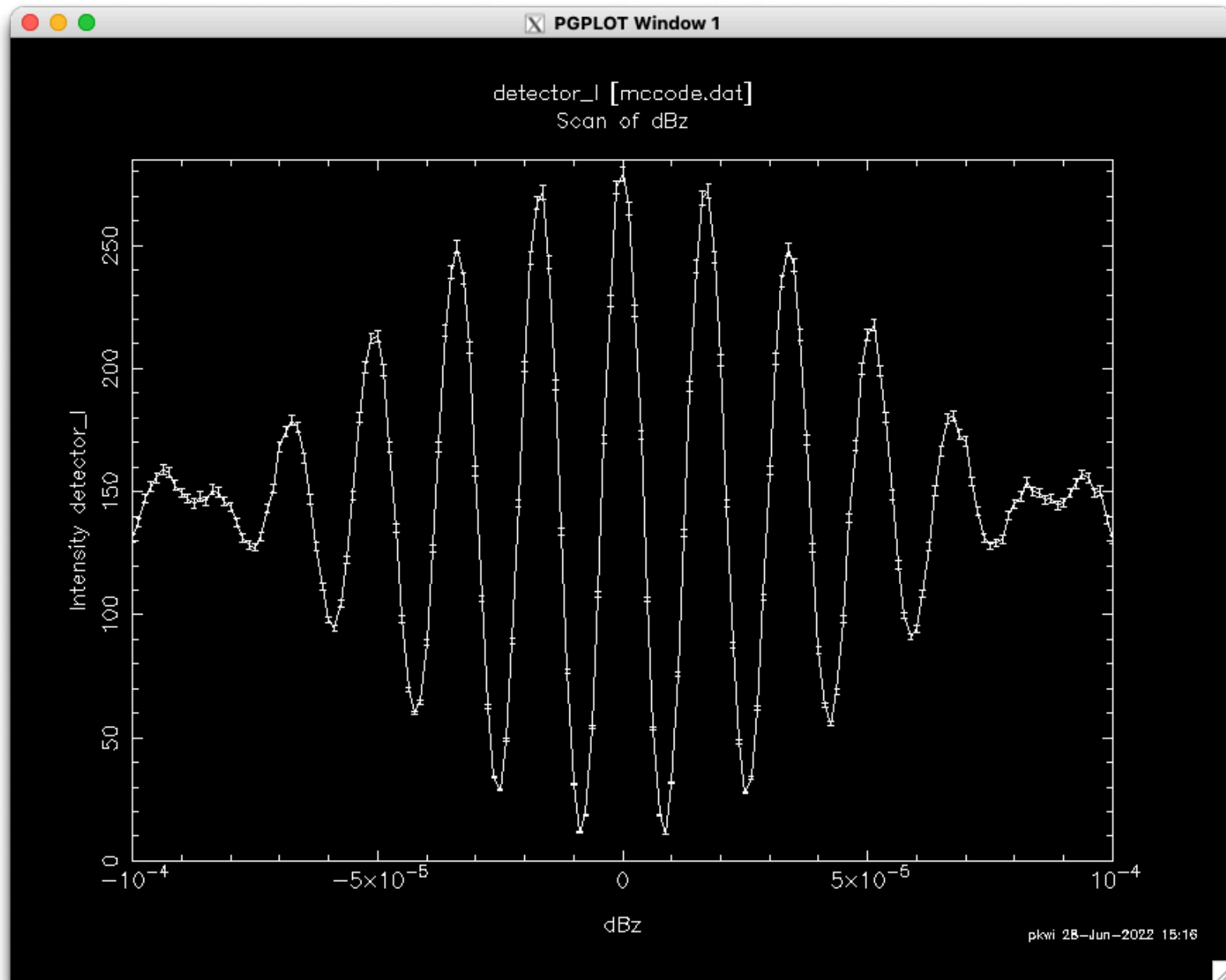
McStas: IN20 TAS model – comparing signals from Vanadium



Generic spin-echo SE_example2.instr

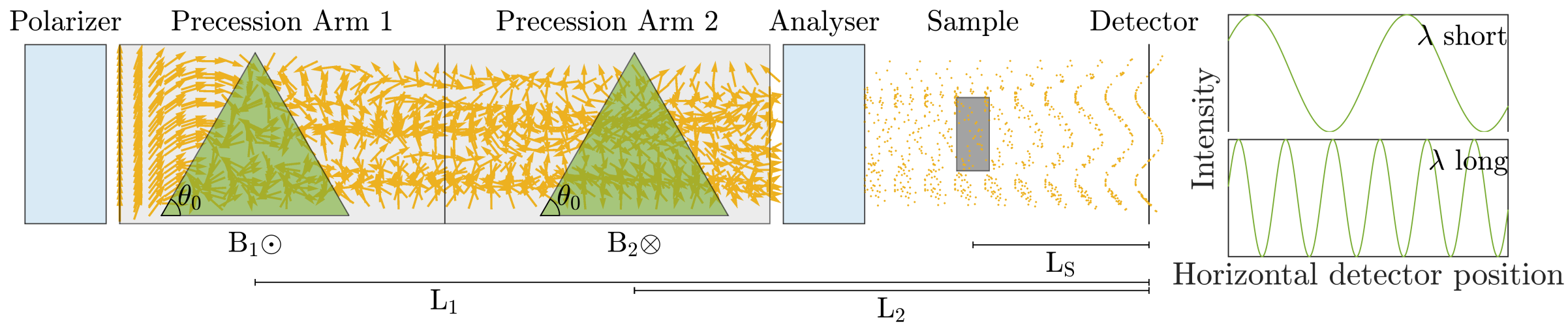


Tuning of guide-field 1 around its value allows us to see spin-echo:

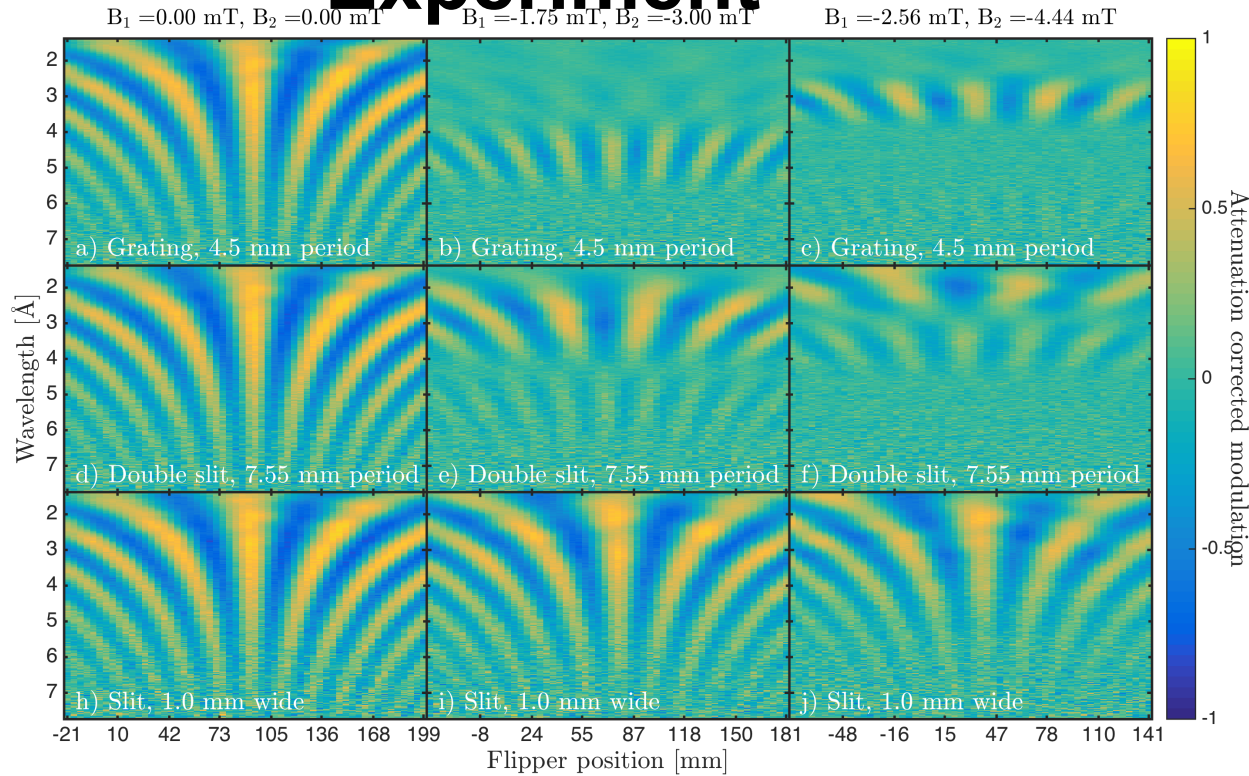


Highlight: McStas example SEMSANS

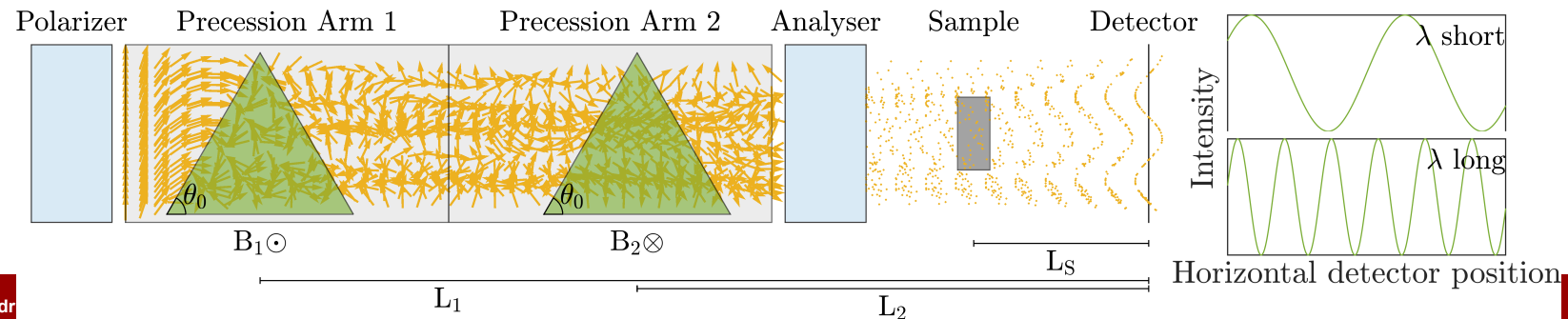
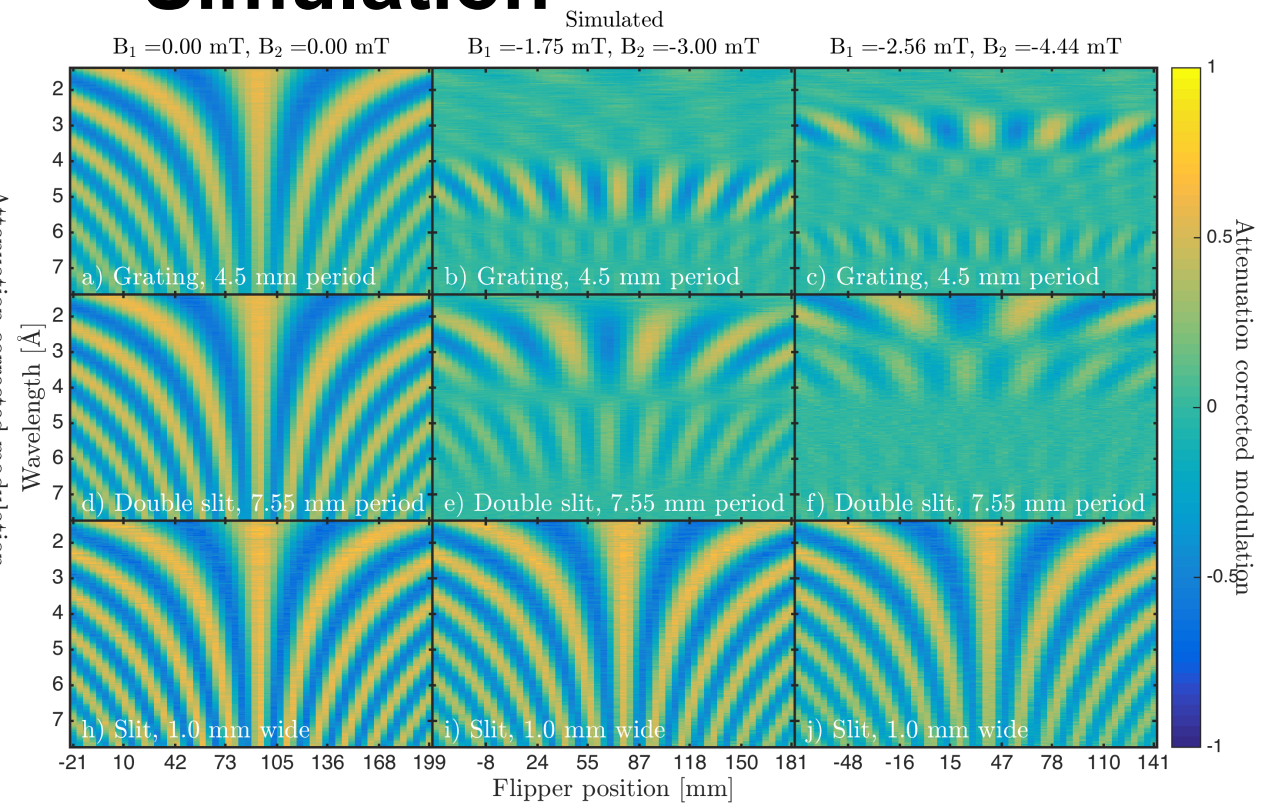
Courtesy: M. Sales et.al.



Experiment



Simulation



- Polarisation docs in manual are **out of date** wrt. the current code/algorithm...
- For now, **don't combine gravity and polarisation!**
(Guide_gravity and Elliptic_guide_gravity can currently not be used within field regions.)
- Pol_Bfield placement can be confusing depending on choice of geometry-parts...
- McStas 3.x does not support
 - User-defined field functions
(No function pointers on GPU... code-generator may rescue us later)
 - Having optics within **tabled** fields
(tabled fields are for now only available in Pol_tabled_field which can not work in "concentric geometries")
- GitHub issues exist and these things will eventually all be fixed!